

UM TAGUM COLLEGE
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
Department of Computing Education
Visayan Village, Tagum City, Philippines



Course Requirements

in

CSE 10/L

CS Professional Elective 2

**Simulation of a Coffee Shop POS System for Faster Order
Processing for Catsy Coffee**

Submitted to:

IRIS MAE C. MENDOZA, MIT

Submitted by:

ASEBEROS, DOMINCE A.
LANZADERAS, MARIEL P.
OBUZA, MA. FE RIBECCA B.

June 17, 2026

1. Introduction of the Project

Project Title:

Simulation of a Coffee Shop POS System for Faster Order Processing for Catsy Cafe

Background:

Catsy Coffee is a friendly neighborhood café located at Purok A, Suaybaguio Street, Barangay Magugpo North, Tagum City. It offers a wide range of beverages and trendy food options that appeal to today's tastes. And currently, the café uses a manual process for taking orders, recording sales, and managing inventory which leads to inefficiency during peak hours.

This project focuses on simulating how a coffee shop handles customer orders using its point-of-sale (POS) system. In a typical setup, customers arrive, line up, place their orders, pay, and then wait for their drinks. Although the process appears simple, delays begin to occur as customer volume increases, especially during peak hours.

This project aims to recreate that flow using Discrete Event Simulation (DES) to observe how the system behaves under different conditions. Instead of relying on assumptions, the simulation will show how customers move through the system and where delays are most likely to occur. Based on these observations, possible improvements, such as adding staff or adjusting service times, can be tested to evaluate their impact on overall performance.

Importance of the Study:

Our project is important because it addresses the operational bottlenecks, long queues, and peak-hour delays that lead to extended customer waiting times and lost sales from customer balking. By utilizing Discrete Event Simulation (DES) rather than relying on assumptions, our study provides a data-driven approach to observe system behavior, locate specific points of congestion, and test potential improvements such as adjusted staffing configurations or system optimizations.

Ultimately, resolving these efficiency issues is crucial for enhancing the overall customer experience, supporting smoother day-to-day operations, increasing resource productivity, and providing solid, data-backed recommendations for smart resource allocation.

2. Problem Definition

The primary problem observed in the system is the occurrence of long queues and delays during peak hours, which happen when specific parts of the workflow—such as cashiers taking longer to process orders or baristas struggling to keep up with drink

preparation—become overloaded and slow down the entire service flow. Addressing this issue is highly important because extended waiting times diminish customer satisfaction and directly result in lost revenue when customers choose to leave rather than wait in line.

Furthermore, improving service speed is vital for supporting smoother day-to-day operations and increasing overall productivity. To resolve these inefficiencies, the simulation aims to analyze how the coffee shop can reduce customer waiting times and maximize efficiency using its existing POS system and resources. Instead of relying on assumptions, the simulation will recreate the customer flow to pinpoint exact bottlenecks and test potential improvements, ultimately providing data-driven recommendations for staffing adjustments, process enhancements, and overall system optimization

3. Objectives of the Simulation

- **Model Customer Flow** – To model and replicate the customer order flow within a coffee shop environment using Discrete Event Simulation (DES).
- **Measure Wait Times** – To measure the average customer's waiting time at each stage of the process, with a specific focus on the cashier and barista stations.
- **Identify Bottlenecks** – To pinpoint bottlenecks and areas of congestion that lead to delays in processing orders during both peak and off-peak hours.
- **Evaluate Staffing Adjustments**: To assess how different staffing configurations, such as adding a second cashier or a second barista, impact overall service efficiency.
- **Analyze System Performance** – To analyze system performance under changing customer arrival rates and identify the maximum throughput the current layout can handle.
- **Provide Data-Driven Recommendations** – To deliver data-backed recommendations aimed at optimizing resource allocation and enhancing the overall customer experience.
- **Assess Reservation Impacts** – To evaluate the impact of offering dedicated reservation tables on overall wait times and observe how prioritizing seated capacity influences customer satisfaction.

4. System Model and Design

Entities:

- **Customers** – individuals who arrive at the coffee shop to place and receive orders; each customer represents one simulation entity moving through the system.
- **Orders** – each customer's order that is entered into the POS system and passed to the preparation area.
- **Reservation Customers** – a subset of arriving customers who have pre-booked a table and are guaranteed seating upon completing their order.

Events:

- **Customer Arrival** – the moment a customer enters the coffee shop and joins the

queue.

- **Order Placement** – the event when a customer reaches the cashier and places their order through the POS system.
- **Payment Processing** – the event of completing the transaction at the POS terminal.
- **Order Preparation Start** – the event when the barista begins preparing the drink after receiving the order details.
- **Order Completion** – the event when the drink is ready and handed off to the customer.
- **Customer Departure** – the customer leaves the system after receiving their order.
- **Customer Balking** – the event when an arriving customer sees the queue and decides to leave without ordering.

Resources:

- **Cashier(s)** – staff responsible for receiving orders and processing payments via the POS system.
- **Barista(s)** – staff responsible for preparing drinks after orders are submitted.
- **POS Terminal** – the electronic system used to record, process, and relay orders.
- **Waiting Queue** – the physical or virtual line where customers wait before being served.
- **Regular Tables** – the general seating area available on a first-come, first-served basis for walk-in dine-in customers.
- **Reservation Tables** – a dedicated pool of tables reserved exclusively for customers who booked in advance.

Queues:

- **Cashier Queues** – The physical lines formed in front of each active cashier where customers wait to place orders. The simulation routes customers to the shortest available cashier line.
- **Barista Prep Queue** – A system-level virtual queue of pending orders that the baristas pull from to prepare drinks.
- **Pickup Area Queue** – The space where takeout customers wait for their completed orders.
- **Dine-In Waiting Queue** – Customers who are seated at their tables waiting for their drinks to be brought to them.

State Variables:

- **Global Clock (env.now)** – Discrete event simulation time, driving all scheduled events.
- **Customer Count** – The total number of customers generated and routed through the system.
- **Queue Lengths (cashierQueueLens)** – Tracks the real-time number of people waiting at each individual cashier's desk.
- **Table Status (table["status"])** – Tracks whether individual tables are available, reserved, or occupied.
- **Completed Customers** – The running tally of customers who have been successfully served.
- **Lost Customers** – The running tally of customers who balked or reneged.

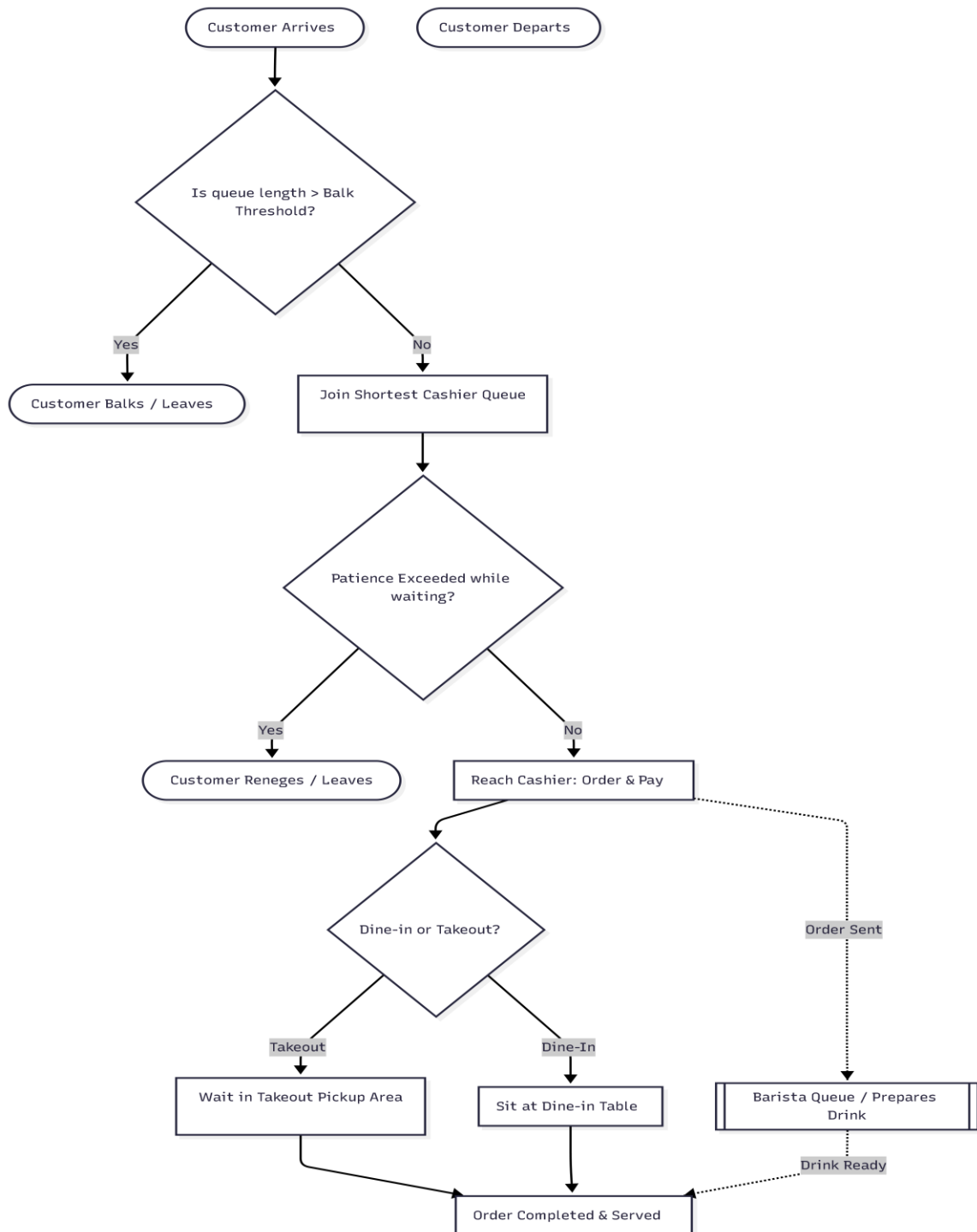


Figure 1. Customer Flow in the Coffee Shop Simulation

5. Assumptions and Input Data

Assumptions Used in the Model

- **Instant Queue Evaluation:** Customers immediately evaluate the queue length upon arrival. If it exceeds their strict personal patience threshold (Balk Threshold), they leave instantly without joining the queue.
- **Decaying Patience:** Customers waiting in the cashier queue possess a finite patience limit (strike system). If they wait too long without moving to the front, their frustration increases until they eventually leave the line (reneging).
- **Pooled Resource Baristas:** Baristas serve orders from a shared, global virtual queue rather than being assigned specifically to a single cashier or order type.
- **Table Allocation:** Customers who cannot find an available table for dine-in will either wait for one or become frustrated. However, Reservation Customers have tables exclusively locked and allocated for them prior to their arrival.

Basis of Arrival Times and Service Times

- **Arrival Times:** Modeled using an exponential distribution based on the configurable average inter-arrival time to simulate realistic, randomized customer foot traffic (Poisson process). Reserved arrivals are deterministic but feature a variable delay (early/late distribution) based on their reserved arrival window limit.
- **Service Times:** Uniform distributions are used for human-driven tasks: deciding the order, paying, and barista preparation times. These are bounded by configurable minimum and maximum second thresholds to accurately account for natural human variation in task completion.

Constraints and Limitations of the Simulation

- **Abstracted Floor Space:** The physical area of the shop is not modeled in square footage; capacity and bottlenecking are strictly enforced by the number of cashiers and tables, not standing room or physical collision.
- **Infinite Inventory:** The simulation assumes that baristas have infinite ingredients. Inventory depletion and restocking delays are outside the scope of the current model.
- **Instantaneous Transit:** Customer physical movement speed is abstracted. The time taken to walk between the entrance, cashier, pickup area, and tables is instantaneous in the logical backend, relying solely on wait/service timers to represent duration.

6. Simulation Implementation

(Note: A live software demonstration of the system—displaying real-time event processing, visual queue handling, time progression, interactive variable adjustment, and dynamic dashboard output generation—is provided separately via the web interface).

7. Experimental Results and Analysis

To evaluate the efficiency of the system, we ran comparative tests between two distinct operational configurations during a simulated peak hour.

Configuration A: Baseline (1 Cashier, 2 Baristas)

This scenario represents a typical understaffed peak hour rush.

- **Average Waiting Time:** Customers experienced severe delays. The time spent in the cashier's queue skyrocketed because the single cashier could not process orders as fast as customers arrived.
- **Queue Length:** The physical line quickly exceeded 8 to 10 people.
- **Throughput:** Extremely poor. Due to the massive queue, the customer balking rate spiked, resulting in lost sales. Only a fraction of the arriving customers actually completed an order.
- **Resource Utilization:** The Cashier was at 100% utilization (overwhelmed), while the Baristas were under-utilized because orders were not reaching them fast enough.
- **Processing Time:** While the actual time to make a drink remained constant, the total cycle time for a customer stretched intolerably long due to queue stagnation.

Configuration B: Optimized (2 Cashiers, 2 Baristas)

This scenario tested the addition of just one extra cashier to combat the bottleneck.

- **Average Waiting Time:** Waiting time dropped significantly. Customers were processed almost as soon as they arrived.
- **Queue Length:** The cashier queue rarely exceeded 2 or 3 people, keeping it well below the average customer's balking threshold.
- **Throughput:** Excellent. Almost all arriving customers were successfully captured and served. Customer balking dropped to near zero.
- **Resource Utilization:** Workload became highly balanced. Both cashiers maintained a steady, manageable flow, and the baristas received a constant, continuous stream of orders, maximizing their efficiency without being starved for work.
- **Processing Time:** Total cycle time was optimized. The bottleneck shifted slightly toward the baristas and the pickup area, but the system overall proved much more capable of handling the high arrival rate.

8. Conclusion and Recommendations

Key Findings of the Simulation

- **Peak Hour Bottlenecks:** The simulation successfully modeled and revealed that during high-traffic periods, the cashier station becomes the primary bottleneck. As order placement times vary, queues build up rapidly.
- **Customer Balking/Reneging:** Extended queue lengths directly trigger a significant increase in lost customers. When the physical line exceeds a customer's balk threshold, or wait times exceed their patience (reneging), revenue is irretrievably lost.
- **Barista Utilization:** The simulation demonstrated that baristas acting as pooled resources handle drink preparation efficiently, provided the cashiers can feed them orders at a steady rate. However, if the cashier bottleneck is resolved without scaling baristas, the bottleneck simply shifts to the pickup area.

Insights Gained from the Analysis

- **System Interdependency:** A coffee shop's throughput is highly interdependent. Speeding up the point-of-sale (POS) interaction alone is insufficient if the preparation queue (baristas) becomes overwhelmed, proving that systemic balance is required.

- **Impact of Reservations:** Customers who utilize the reservation system maintain a significantly higher satisfaction and throughput rate because they bypass the most volatile part of the queuing system (the cashier waiting line).
- **Data-Driven Staffing:** Using DES eliminates guesswork. We observed that adding just one extra cashier during peak bursts drastically reduced average queue length and almost eliminated customer balking.

Recommendations for Improving the System

1. **Dynamic Staff Reallocation:** Cross-train employees to dynamically open a secondary POS terminal (cashier) when the queue length exceeds 5 customers. This prevents the immediate onset of balking.
2. **Order Pre-Processing:** Introduce a mobile-ordering or kiosk system to reduce the time spent "deciding and paying" at the cashier, shifting the bottleneck away from human interaction at the POS terminal.
3. **Promote the Reservation System:** Encourage customers to utilize the table reservation system, as the simulation proves it smooths out arrival spikes and guarantees service without relying on the unpredictable walk-in queue.

Appendix: Core Code Snippets

The following code snippets highlight the core Discrete Event Simulation logic, event-driven data structures, and mathematical modeling used in the Python backend.

1. Defining Simulation Resources & Data Emitter (cafe_env.py)

This snippet demonstrates how the `simpy` library is used to create constrained capacity resources (Cashiers and Baristas). It also shows how simulation events are structured into JSON data payloads containing the exact timestamp, event type, and customer ID.

```
class CafeEnvironment:
    def __init__(self, env, event_queue, config):
        self.env = env
        self.event_queue = event_queue
        self.config = config

        # Define capacity-constrained resources
        self.cashiers = [simpy.Resource(env, capacity=1) for _ in
range(config["cashier_count"])]
        self.baristas = simpy.Store(env,
capacity=config["barista_count"])
        for i in range(config["barista_count"]):
            self.baristas.put(i)

    def emit(self, event_type, customer_id, extra=None):
        """Constructs and pushes data events to the frontend UI"""
        payload = {
            "time": round(self.env.now, 2),
            "event": event_type,
            "customer_id": customer_id
        }
        if extra: payload.update(extra)
        self.event_queue.put_nowait(payload)
```

2. Customer Event Flow & Balking Logic (customer_flows.py)

This snippet outlines the step-by-step logic of a customer entity flowing through the system. It highlights the mathematical evaluation of queue lengths against the customer's threshold, which dictates whether they join the queue or "balk" (leave the store).

```
def customer_process(env, cafe, name, config, force_takeout=False,
is_reservation=False):
    cafe.emit("arrive", name)

    # 1. Balking Logic (Data check)
    shortest_queue_idx, shortest_len = get_shortest_cashier_queue(cafe)
    balk_threshold = config.get("balk_threshold", 8)

    if shortest_len >= balk_threshold and random.random() <
config.get("balk_prob", 0.5):
        cafe.emit("balking_start", name)
        yield env.timeout(random.uniform(1.0, 3.0)) # Consideration
delay
        cafe.emit("balk_leave", name)
        cafe.lost_customers += 1
        return

    # 2. Join Queue & Await Cashier
    cashier_res = cafe.cashiers[shortest_queue_idx]
    cafe.emit("queue_cashier", name, {"cashier_index":
shortest_queue_idx})

    with cashier_res.request() as req:
        yield req

    # 3. Order Processing Event
    cafe.emit("start_deciding", name, {"cashier_index":
shortest_queue_idx})
    yield env.timeout(random.uniform(config["decide_min"],
config["decide_max"]))
```